

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Unit Test 1

Unit Test 1

Class: Class 10

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Networking Concepts

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

- a. Which network topology requires a central hub or switch?
 - b. Bus topology
 - c. Star topology
 - d. Ring topology
 - e. Mesh topology
- b. The OSI model has how many layers?
 - c. 5
 - d. 6
 - e. 7
 - f. 8
- c. Which of the following is a Class B IP address?
 - d. 192.168.1.1
 - e. 172.16.0.1
 - f. 10.0.0.1
 - g. 127.0.0.1
- d. Which protocol is used to send email?
 - e. HTTP
 - f. FTP
 - g. SMTP
 - h. POP3
- e. What is the purpose of a firewall?
 - f. To boost internet speed
 - g. To filter incoming and outgoing network traffic
 - h. To store backup data
 - i. To manage user accounts

Section II: Short Answer (5 x 1 = 5 marks)

- a. Explain the difference between **IPv4** and **IPv6** addressing. Why is IPv6 needed?
- b. Draw a neat labelled diagram of **Star Topology**. List two advantages and two disadvantages.
- c. Write the full forms and explain the purpose of: HTTP, FTP, SMTP, DNS, TCP.
- d. What is a **MAC address**? How is it different from an **IP address**?
- e. Explain how data travels from one computer to another through the Internet, mentioning at least 3 devices/protocols involved.